MEATU DISTRICT COUNCIL, FORM THREE MOCK EXAMINATION OCTOBER 2019.

033/1 BIOLOGY MARKING SCHEME

SECTION A (15 Marks)

1. Multiple choice 10marks 1 @ each

Question number	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
Answer	C	A	A	E	D	B	D	D	B	E

2. Matching items 5marks 1 @ each

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B	D	E	A	C	B

SECTION B (60Marks)

3. (a) Waste disposal - Is a way/ways of eliminating or removing waste from the environment.

OR

Waste disposal - Is the act of getting rid of unwanted items and materials with no value in excess of their basic material value.

(b) Waste disposal is a problem because of the following challenges:-

(i) peoples bad habits
(ii) Use of excessive packaging materials
(iii) The failure by cleaners in towns to get rid of wastes in public areas.
(iv) poor or inadequate disposal facilities
(v) Failure by the local authorities to enforce effective punishment on those who litter public
(vi) Improper management of wastes

Total Marks 6 (a) 2 marks
(b) 4 marks (four points only)

4. (a) Adaptive features of the mammalian heart to its functions are:-

(i) It has muscular walls

(ii) It has cardiac muscles

(iii) It has valves

(iv) The heart have a septum which separates oxygenated blood from deoxygenated blood.

(v) The heart is connected to large blood vessels

(vi) It has coronary artery and coronary vein

(b) Precautions to be taken during blood transfusion are:-

(i) To check blood compatibility between the donor and recipient.

(ii) The donors blood should be screened to ensure that it does not have pathogens.

(iii) The blood should be taken from a vein, probably in the arm.

(iv) An anticoagulant should be put in the blood to prevent clotting and then stored above freezing point.

(v) The donor should be aged 18–60 years and healthy with 50kilogram of weight or above.

(vi) Donated blood should be kept in a refrigerator

Total marks 6 (a) 3marks (3points)

(b) 3marks (3points)

5. (a) Transpiration - Is the process by which plants lose excess water in the form of water vapour to the atmosphere.

WHILE

Guttation - Is the process by which plants lose water as droplets through special glands found where veins are in contact with the leaf margin.

5. (b) Significances of transpiration are;-

(i) It helps to cool the plants

(ii) Transpiration enables the loss of excess water from the plants

(iii) It helps to maintain transpirational pull which enables absorption and distribution of water and mineral salts.

Total marks 6 (a) 3 (1.5 per each)
(b) 3 (1 per each point)

6. (a) Importance of movement in plants and animals are;-

(i) Enables animals to find habitats

(ii) It enables plants to disperse seeds for sustainable generations

(iii) Movement enables animals to find food

(iv) It enables plants to reproduce sexually (Flowering plants) through pollination

(v) Movement enables animals to move away from predators.

(v) It also enable animals to come together for reproductive purposes.

(b) Functions of the human skeleton are;-

(i) support the body

(ii) protect delicate organs like (brain, heart, lungs, liver, spinal cord, etc).

(iii) It gives shape of the body of an organism

(iv) synthesis of white and red blood cells.

(v) Enable locomotion and movement

(vi) Provide surface area for muscle attachment (anchorage)

Total marks 6 (a) 3 marks (3 points)
(b) 3 marks (3 points)

7. (a) Actions which take place in the human body in response to low temperature in the surroundings are;
(i) Shivering
(ii) Increase in metabolic rate
(iii) Vasoconstriction of arterioles
(iv) Contraction of hair erector muscles

(b) Actions which take place in the human body in response to rise in body temperature are:-
(i) Vasodilation
(ii) Decrease in metabolic rate
(iii) Sweating
(iv) Relaxation of hair erector muscles

Total marks 6 (a) 3 marks (3 points)
(b) 3 marks (3 points)

8. (a) (i) Breathing - Is the process of drawing air into and out of the lungs.

BUT

Respiration - Is the process by which food substances are chemically broken down in the living cells to release energy.

(ii) Inhalation - Is the process of drawing air into the lungs

WHEREBY

Exhalation - Is the process of drawing air out of the lungs.

(b) It is healthier to breath through the nose than the mouth because in the nose there is mucus and hairs (cilia) which trap dust and pathogens (microorganisms).

Total marks 6 (a) (i) 2 marks
(ii) 2 marks
(b) 2 marks

9. (a) Two types of blood cells are:-
(i) Red blood cells
(ii) White blood cells

(b) Differences between arteries and veins

Arteries	Veins
(i) They carry oxygenated blood except pulmonary artery	They carry deoxygenated blood except pulmonary vein
(ii) They have thick and muscular walls	They have thin and less muscular walls
(iii) They carry blood away from the heart	They carry blood towards the heart
(iv) Blood flows under high pressure	Blood flows under low pressure
(v) They have a small lumen	They have a wider lumen
(vi) They are deep rooted	They are superficial
(vii) Blood flows in pulses	Blood flows smoothly.
(viii) They have no valves except at the base of aorta and pulmonary artery	They have valves at intervals, which prevent backward flow of blood.

Total marks 6 (a) 2 marks
(b) 4 marks (4 points only)

10. (a) Types of muscles found in mammals are:-
(i) Cardiac muscles
(ii) Skeletal muscles
(iii) Smooth muscles

10. (b) Adaptations of muscles to their roles are.

(i) They are elastic (They have elasticity tendency)

(ii) They are supplied with blood for easy transportation of materials

(iii) Muscles have numerous mitochondria which produce enough energy.

(iv) They are able to store calcium which facilitate relaxation and contraction easily.

(v) skeletal muscles have tendons which enable them to be attached on bones easily.

Total marks 6 (a) 3 marks
(b) 3 marks (3 points)

11. (a) (i) Tissue - Is a group of similar cells forming a functional unit.

(ii) Animal tissues are epithelial tissue, blood tissue, muscular tissue, nerve tissue, bone tissue, supporting tissue, etc
Plant tissues are meristematic tissue, parenchyma tissue, collenchyma tissue, xylem tissue, xylem tissue

(b) Differences between plant cell and animal cell

Plant cell	Animal cell
(i) Have a cell wall of cellulose	Have no cell wall
(ii) Have chloroplasts with chlorophyll	Have no chloroplasts
(iii) Have a definite shape	Have no definite shape
(iv) Have large permanent vacuoles	Have small temporary vacuoles or none.
(v) Autotrophic nutrition	Heterotrophic nutrition

11. (b)

	Plant cell	Animal cell
(vi)	store food in the form of starch grains.	store food in the form of glycogen.
(vii)	They usually lack centrioles.	Have centrioles and normally exist in pairs.

Total marks 6 (a) (i) 1 mark
(ii) 3 mark (3 points)
(b) 2 marks (only 2 points)

12. (a) structures used for locomotion in

	Animal	structure for locomotion
(i)	Amoeba	Pseudopodia
(ii)	Fish	Fins
(iii)	Paramecium	Cilia

(b) Movement - Is the change of position and posture WHILE Locomotion - Is the change of position of the whole body of an organism from one place to another.

Total marks 6 (a) 3 marks (1 mark each)
(b) 3 marks (1.5 mark each)

SECTION C (25 marks)

13. (a) (i) Medulla oblongata controls all unconscious activities of the body such as breathing, heartbeat, digestion, dilation and contraction of blood vessels, temperature regulation, secretion of juices from glands, etc.

(ii) Hypothalamus - It is involved in regulation of body temperature (thermoregulation)

(iii) Cerebellum - It regulates and coordinates body movements which are concerned with equilibrium.

(iv) Cerebrum - It controls voluntary (conscious) activities like speech, reasoning memory, decision making behaviour and imagination. The cerebrum interlinks impulses and coordinates responses. It also controls activities such as running, walking, eating and playing.

(b) (i) Drug abuse - Is the use of drugs for the purposes of self-indulgence.

(ii) Effects of drug abuse are

Loss of esteem by the users as they may be rejected by friends and family.
Loss of work hours by society as many users take time off to recover from the side-effects of their habit.
Users may lose their jobs as a result of repeated failure to turn up for work
Users often turn to crime to fund their habit
relationship may break up
Increases of street children
It may lead to mental confusion or death
Easy spread of sexually transmitted disease
Famimily disintergration

13. (b) (iii) control measures against drug abuse are:-

provission of education
Enaction of laws and regulations
Keeping away from peer pressure
Engaging in physical aitivities and physical exercise.
Guidance and counselling
Medical advice and intervention
Teaching of good morals at a family level.

Total marks 15

(a) (i) – (iv) = 4 marks
(b) (i) = 1 mark
(ii) = 6 marks (3 points)
(iii) = 4 marks (2 points)

14. Malaria is a protozoan disease caused by plasmodia (singular plasmodium).

Transmission of malaria

The plasmodium species is transmitted from a person to a person by the female anopheles mosquitoes through blood consumption. Also in a rare situations, transmission may occur through transfusion of infected blood by malaria parasites from a sick person to a healthy person.

Symptoms of malaria

Chills and high fever which may be associated with sweating.
Nausea and vomiting
Loss of appetite
Frontal headache and pain in the joint
General body weakness

14. Preventive measures of malaria

Draining away of unwanted ponds, pools and swamps. If possible stagnant water and sluggish rivers should be turned into swifter streams.
Use of insecticides to kill mosquitoes by spraying to areas with mosquitoes and their breeding sites.
Use of mosquito nets during the night
Screen the windows with mesquito proof wire mesh.
cutting off (slashing) long grasses around our homes to eradicate mesquitoes
Use of biological control.
Take preventive medicines regularly.

15. Mechanism of breathing in human being

Breathing is the processes of drawing air into and out of the lungs. The mechanism of breathing can be described into two processes namely inhalation (inspiration) and exhalation (expiration).

Inhalation (inspiration) is the process of drawing air into the lungs. Also exhalation (expiration) is the process of drawing air out of the lungs.

⇒ During inspiration and expiration the following events occur as per table below:-

Inhalation	Exhalation
External intercostal muscles contract	The external intercostal muscles relax
Internal intercostal muscles relax	Internal intercostal muscles contract

15.

Inhalation	Exhalation
The ribcage is lifted outwards and upwards	The ribcage moves inwards and downwards.
The diaphragm contracts and flattens	The diaphragm relaxes and becomes dome-shaped
The volume of the thoracic cavity increases as the pressure decreases	The volume of the thoracic cavity decreases as pressure increases. This forces air out of the thoracic cavity.
Air enters the alveoli through the nostril, pharynx, glottis, trachea, bronchioles and finally alveoli	Air leaves the alveoli through the bronchioles, trachea, glottis, pharynx and finally nostrils.

Marks 10 (Either question number 14 or 15).

On 14

Introduction = 2 marks
Main body = 7 marks
conclusion = 1 mark

On 15

Introduction = 3 marks
Main body = 6 marks
Conclusion = 1 mark